

# Notes: Rock (JFE, 1986)

## Why New Issues Are Underpriced

### Contents

|           |                                                                  |          |
|-----------|------------------------------------------------------------------|----------|
| <b>1</b>  | <b>Big picture</b>                                               | <b>3</b> |
| <b>2</b>  | <b>Incremental contribution of the paper</b>                     | <b>3</b> |
| <b>3</b>  | <b>Institutional setting: firm-commitment IPOs</b>               | <b>4</b> |
| <b>4</b>  | <b>Model primitives</b>                                          | <b>4</b> |
| 4.1       | Assets, issuer, and investors . . . . .                          | 4        |
| 4.2       | Information assumptions . . . . .                                | 5        |
| <b>5</b>  | <b>Informed and uninformed demand</b>                            | <b>5</b> |
| 5.1       | Informed demand . . . . .                                        | 5        |
| 5.2       | Uninformed demand . . . . .                                      | 5        |
| 5.3       | Aggregate demand by state . . . . .                              | 5        |
| <b>6</b>  | <b>Allocation and rationing probabilities</b>                    | <b>6</b> |
| 6.1       | Allocation mechanism . . . . .                                   | 6        |
| 6.2       | Derivation of rationing probabilities . . . . .                  | 6        |
| <b>7</b>  | <b>Uninformed investor problem</b>                               | <b>6</b> |
| 7.1       | Terminal wealth . . . . .                                        | 6        |
| 7.2       | Expected utility . . . . .                                       | 6        |
| 7.3       | First-order condition . . . . .                                  | 7        |
| <b>8</b>  | <b>Rational-expectations equilibrium</b>                         | <b>7</b> |
| 8.1       | Equilibrium system . . . . .                                     | 7        |
| 8.2       | Why the issuer needs uninformed investors . . . . .              | 7        |
| <b>9</b>  | <b>Zero-demand probability and large-market logic</b>            | <b>7</b> |
| 9.1       | Definition . . . . .                                             | 7        |
| 9.2       | Interpretation of $b_0(p)$ . . . . .                             | 8        |
| 9.3       | Theorem 1 . . . . .                                              | 8        |
| 9.4       | Large-market convergence . . . . .                               | 8        |
| <b>10</b> | <b>Demand response and the issuer's opportunity set</b>          | <b>8</b> |
| 10.1      | Demand identity . . . . .                                        | 8        |
| 10.2      | Theorem 2 . . . . .                                              | 9        |
| 10.3      | Full-subscription price . . . . .                                | 9        |
| <b>11</b> | <b>Example: uniform values and quadratic utility</b>             | <b>9</b> |
| 11.1      | Assumptions . . . . .                                            | 9        |
| 11.2      | Uninformed demand . . . . .                                      | 10       |
| 11.3      | Zero-demand probability under the uniform distribution . . . . . | 10       |
| 11.4      | Full-subscription price in the large-market example . . . . .    | 10       |

|           |                                                                          |           |
|-----------|--------------------------------------------------------------------------|-----------|
| 11.5      | Finite-market comparison . . . . .                                       | 10        |
| <b>12</b> | <b>Multiple allocation equilibria and hot/cold markets</b>               | <b>11</b> |
| 12.1      | Relative allocation probability . . . . .                                | 11        |
| 12.2      | Demand explosion below full subscription . . . . .                       | 11        |
| 12.3      | Hot and cold issue interpretation . . . . .                              | 11        |
| <b>13</b> | <b>Issuer's optimal offer price</b>                                      | <b>11</b> |
| 13.1      | Issuer objective . . . . .                                               | 11        |
| 13.2      | Issuer terminal wealth . . . . .                                         | 11        |
| 13.3      | Why the issuer will not choose zero uninformed demand . . . . .          | 12        |
| 13.4      | Optimal price in the example . . . . .                                   | 12        |
| <b>14</b> | <b>Empirical implications and tests</b>                                  | <b>12</b> |
| <b>15</b> | <b>Tables and figures</b>                                                | <b>13</b> |
| 15.1      | Table 1: Terminal wealth of the uninformed investor . . . . .            | 13        |
| 15.2      | Table 2: Uninformed demand and allocation probabilities . . . . .        | 13        |
| 15.3      | Figure 1: Probability of receiving underpriced shares . . . . .          | 14        |
| 15.4      | Figure 2: Relative allocation probability and multiple regimes . . . . . | 14        |
| 15.5      | Table 3: Issuer terminal wealth . . . . .                                | 14        |
| <b>16</b> | <b>Short summary</b>                                                     | <b>15</b> |
| <b>17</b> | <b>Conclusion</b>                                                        | <b>16</b> |
| 17.1      | Core conclusion . . . . .                                                | 16        |
| 17.2      | Economic intuition . . . . .                                             | 16        |
| 17.3      | Incremental contribution revisited . . . . .                             | 16        |
| 17.4      | Implications for issuers and underwriters . . . . .                      | 16        |
| 17.5      | Empirical implications . . . . .                                         | 16        |
| 17.6      | Limitations . . . . .                                                    | 16        |
| 17.7      | Takeaway . . . . .                                                       | 16        |

## 1 Big picture

- **Question.** Why are initial public offerings (IPOs), especially firm-commitment offerings, systematically priced below their subsequent aftermarket value?
- **Core answer.** Underpricing is compensation to uninformed investors for an adverse allocation problem. Informed investors crowd into good IPOs and withdraw from bad IPOs, so uninformed investors receive too few good issues and too many bad issues.
- **Key economic friction.** The issuer and uninformed investors cannot observe aggregate informed demand before the offering price is set and allocations are made. Price is fixed before the market clears, so price cannot fully reveal informed demand.
- **Main mechanism.** If the offer price equals expected value, informed investors submit large orders only when the issue is good. Good issues become oversubscribed and rationed. Bad issues are not oversubscribed and uninformed investors receive full allocations. Conditional on receiving shares, uninformed investors face a winner's curse.
- **Equilibrium discount.** The issuer must set the offer price below expected value so that uninformed investors are willing to participate despite biased rationing.
- **Issuer trade-off.** Lower prices raise uninformed participation and improve the probability that the issue is fully sold, but lower prices reduce expected proceeds when the issue would have sold anyway.
- **Important theoretical insight.** Rationing itself is not the problem. Biased rationing is the problem: the probability of receiving good shares is lower than the probability of receiving bad shares.
- **Main empirical implication.** Weighting IPO returns by actual allocation probabilities should remove abnormal profits for uninformed investors. Underpricing in posted IPO returns does not imply that ordinary investors can earn those returns after rationing.
- **Economic takeaway.** IPO underpricing is not a mistake or a mystery in this model. It is the equilibrium price of attracting uninformed capital into a market where informed demand selectively appears in good states.

## 2 Incremental contribution of the paper

- **From empirical anomaly to microfoundation.** Earlier evidence documented positive first-day or short-run IPO returns. Rock provides a formal mechanism for why IPOs must be underpriced in equilibrium.
- **Winner's curse in IPO allocations.** The paper adapts auction-style adverse selection logic to firm-commitment IPOs: uninformed investors condition on receiving an allocation, and that event is bad news.
- **Asymmetric information among investors, not just issuer-underwriter agency.** Unlike explanations based on an informed underwriter or issuer, the model emphasizes informational asymmetry among investors: some investors know more about true value than others.

- **Endogenous uninformed participation.** The issuer needs uninformed investors because informed investors do not buy enough in all states. Underpricing is the price of keeping uninformed investors in the market.
- **Allocation probability as a state-price wedge.** The model shows that the expected payoff to an uninformed investor depends on the relative rationing probability  $b/b'$ , not simply on expected aftermarket returns.
- **Large-market simplification.** In large markets, uninformed investors demand approximately zero abnormal profit. This produces a simple zero-demand probability  $b_0(p)$  that characterizes equilibrium rationing.
- **Full-subscription price.** The paper distinguishes merely attracting some uninformed demand from attracting enough uninformed demand to guarantee full sale even in bad states.
- **Hot and cold issue interpretation.** The example suggests that multiple demand/allocation equilibria at similar prices may be related to hot versus cold IPO markets.
- **Testable allocation implication.** The model implies that raw first-day returns overstate the return available to uninformed investors because the best issues are more heavily rationed.

### 3 Institutional setting: firm-commitment IPOs

- In a firm-commitment IPO, the issuer and investment bank set a fixed offer price  $p$  and a fixed quantity  $Z$  of shares before the offering clears.
- After price is fixed, investors submit indications or orders. The issuer cannot continuously update price as in a standard auction.
- If demand exceeds shares offered, the underwriter rations the shares. If demand is below supply, some shares remain unsold and are typically absorbed or disposed of later.
- The key informational event is whether the issue is oversubscribed. This is not fully observed until after the offering date.
- The model treats IPO selling as auction-like because investors bid for shares, but not as a standard auction because the price is set by the issuer rather than by investor bids.

**Interpretation.** The institutional timing is essential. If price adjusted to demand before allocation, informed demand would be partly revealed and the adverse allocation problem would be reduced. In a firm-commitment IPO, price is fixed before aggregate demand is observed, so informed investors can exploit their information by applying only to attractive issues.

## 4 Model primitives

### 4.1 Assets, issuer, and investors

- There are two assets: a safe asset with gross return normalized to 1 and a risky new issue.
- The aftermarket value per share of the IPO is denoted  $\tilde{v}$ .
- The unconditional mean value per share is  $\bar{v} = \mathbb{E}[\tilde{v}]$ .

- The issuer chooses an offer price  $p$  and offers  $Z$  shares.
- There are  $N$  uninformed investors, each with wealth normalized to 1.
- There is a group of informed investors. When the issue is underpriced, their aggregate dollar demand is  $I$ .

## 4.2 Information assumptions

- **Informed investors know  $\tilde{v}$ .** They can identify whether the IPO is worth buying at price  $p$ .
- **Uninformed investors only know the distribution of  $\tilde{v}$ .** They cannot distinguish good and bad issues before submitting orders.
- **The issuer is also treated as uninformed relative to the market as a whole.** The issuer and underwriter may know much about the firm, but they do not know the full aggregate information held by all market participants.
- **Informed investors cannot short sell IPO shares and cannot sell their private information.** They only profit by submitting purchase orders in good states.
- **Informed demand is bounded:**  $I \leq \bar{v}Z$ .

**Interpretation.** The assumption that the issuer is uninformed should not be read literally as ignorance about its own firm. It means that the market's pooled information about valuation, discount rates, substitutes, competitors, and demand may dominate the issuer's information.

## 5 Informed and uninformed demand

### 5.1 Informed demand

Informed investors buy only if the issue is underpriced:

$$D^I(p, \tilde{v}) = \begin{cases} I, & p < \tilde{v}, \\ 0, & p > \tilde{v}. \end{cases} \quad (1)$$

The case  $p = \tilde{v}$  is immaterial under a continuous distribution.

### 5.2 Uninformed demand

Each uninformed investor chooses a fraction  $T$  of wealth to submit to the offering. Because investors cannot short sell IPO shares,

$$T^* = \max\{0, T\}. \quad (2)$$

Aggregate uninformed dollar demand is  $NT^*$ .

### 5.3 Aggregate demand by state

The aggregate dollar value of orders is

$$D(p, \tilde{v}) = \begin{cases} NT^* + I, & p < \tilde{v}, \\ NT^*, & p > \tilde{v}. \end{cases} \quad (3)$$

The key asymmetry is that informed demand appears only in good states. Hence good issues are more likely to be rationed.

## 6 Allocation and rationing probabilities

### 6.1 Allocation mechanism

Rock assumes that submitted orders are randomly assigned lottery numbers and filled until either shares or orders are exhausted. This implies that, conditional on the issue state, an order's probability of being filled is independent of order size.

Let

- $b$  be the probability that an uninformed order is filled when  $\tilde{v} > p$ ;
- $b'$  be the probability that an uninformed order is filled when  $\tilde{v} < p$ .

### 6.2 Derivation of rationing probabilities

If  $\tilde{v} > p$ , total dollar orders are  $NT^* + I$ . The dollar value of the issue is  $pZ$ . Ignoring small round-off errors, the allocation probability is

$$b = \min \left\{ \frac{pZ}{NT^* + I}, 1 \right\}. \quad (4)$$

If  $\tilde{v} < p$ , informed investors do not participate, so total dollar orders are  $NT^*$ . The allocation probability is

$$b' = \min \left\{ \frac{pZ}{NT^*}, 1 \right\}. \quad (5)$$

Therefore,

$$b \leq b'. \quad (6)$$

**Economic message.** An allocation is worse news than it looks. If the issue is good, informed demand crowds in and lowers the chance that an uninformed investor receives shares. If the issue is bad, informed demand disappears and the uninformed investor is more likely to receive shares.

## 7 Uninformed investor problem

### 7.1 Terminal wealth

If an uninformed investor orders fraction  $T$  and receives shares, wealth changes from 1 to

$$1 + T \left( \frac{\tilde{v}}{p} - 1 \right). \quad (7)$$

If the investor does not receive an allocation, the order is converted into the safe asset and terminal wealth is 1.

### 7.2 Expected utility

Let  $b_e$  and  $b'_e$  denote the uninformed investor's beliefs about allocation probabilities. Expected utility is

$$\begin{aligned} \mathbb{E}U(W) = & b_e \mathbb{P}(\tilde{v} > p) \mathbb{E} \left[ U \left( 1 + T \left( \frac{\tilde{v}}{p} - 1 \right) \right) \mid \tilde{v} > p \right] \\ & + b'_e \mathbb{P}(\tilde{v} \leq p) \mathbb{E} \left[ U \left( 1 + T \left( \frac{\tilde{v}}{p} - 1 \right) \right) \mid \tilde{v} \leq p \right] \\ & + [1 - b_e \mathbb{P}(\tilde{v} > p) - b'_e \mathbb{P}(\tilde{v} \leq p)] U(1). \end{aligned} \quad (8)$$

### 7.3 First-order condition

For an interior choice of  $T$ , the first-order condition is

$$0 = \frac{b_e}{b'_e} \mathbb{P}(\tilde{v} > p) \mathbb{E} \left[ U' \left( 1 + T \left( \frac{\tilde{v}}{p} - 1 \right) \right) \left( \frac{\tilde{v}}{p} - 1 \right) \mid \tilde{v} > p \right] \\ + \mathbb{P}(\tilde{v} \leq p) \mathbb{E} \left[ U' \left( 1 + T \left( \frac{\tilde{v}}{p} - 1 \right) \right) \left( \frac{\tilde{v}}{p} - 1 \right) \mid \tilde{v} \leq p \right]. \quad (9)$$

**Key implication.** Uninformed demand depends on the *relative* allocation probability  $b_e/b'_e$ . Equal rationing in both states would not change expected returns. The distortion arises because good-state rationing is stronger than bad-state rationing.

## 8 Rational-expectations equilibrium

### 8.1 Equilibrium system

In equilibrium, beliefs must equal the allocation probabilities generated by actual demand. Since uninformed demand depends only on  $b/b'$  and  $p$ , write it as  $T^*(b/b', p)$ . The equilibrium is characterized by

$$b = \min \left\{ \frac{pZ}{NT^*(b/b', p) + I}, 1 \right\}, \quad (10)$$

$$b' = \min \left\{ \frac{pZ}{NT^*(b/b', p)}, 1 \right\}, \quad (11)$$

plus the investor optimality condition.

### 8.2 Why the issuer needs uninformed investors

The informed buy only in good states. If the issuer relies solely on informed investors, it sells only when  $\tilde{v} > p$  and is stuck with the issue when  $\tilde{v} < p$ . This is unattractive when owners are risk averse or when the firm needs reliable funding for investment. Uninformed participation creates a more reliable market for the issue, but uninformed investors must be compensated for adverse allocation.

## 9 Zero-demand probability and large-market logic

### 9.1 Definition

For large markets, each uninformed investor's order is small relative to wealth, so marginal participation requires approximately zero abnormal expected profit. Suppose bad issues are not rationed, so  $b' = 1$ . At the margin  $T = 0$ , the investor is indifferent when

$$b \mathbb{P}(\tilde{v} > p) \mathbb{E}[\tilde{v} - p \mid \tilde{v} > p] = \mathbb{P}(\tilde{v} < p) \mathbb{E}[p - \tilde{v} \mid \tilde{v} < p]. \quad (12)$$

This gives the zero-demand probability:

$$b_0(p) = \frac{\mathbb{P}(\tilde{v} < p) \mathbb{E}[p - \tilde{v} \mid \tilde{v} < p]}{\mathbb{P}(\tilde{v} > p) \mathbb{E}[\tilde{v} - p \mid \tilde{v} > p]}. \quad (13)$$

## 9.2 Interpretation of $b_0(p)$

- $b_0(p)$  is the lowest good-state allocation probability that uninformed investors will tolerate at offer price  $p$ .
- If  $b < b_0(p)$ , good issues are too rationed relative to bad issues, and uninformed investors withdraw.
- If  $b > b_0(p)$ , uninformed investors find participation attractive and submit positive demand.
- In large markets, equilibrium  $b(p, N)$  converges to  $b_0(p)$ .

## 9.3 Theorem 1

For any  $0 < p < \bar{v}$ , the fixed point

$$b = \min \left\{ \frac{pZ}{NT^*(b, p) + I}, 1 \right\} \quad (14)$$

has a unique solution  $b(p, N)$ , and

$$b(p, N) > b_0(p) > 0. \quad (15)$$

**Economic meaning.** The equilibrium allocation probability is pinned down by two opposing forces. A higher  $b$  makes uninformed participation more attractive, which increases demand and pushes actual allocation probabilities down. This monotonic feedback produces a unique fixed point.

## 9.4 Large-market convergence

Rock proves that under smoothness conditions,

$$\lim_{N \rightarrow \infty} b(p, N) = b_0(p), \quad (16)$$

and

$$\lim_{N \rightarrow \infty} \frac{d}{dp} b(p, N) = \frac{d}{dp} b_0(p). \quad (17)$$

This is useful because  $b_0(p)$  depends only on the distribution of  $\tilde{v}$  and the offer price, not on the exact utility function of investors.

# 10 Demand response and the issuer's opportunity set

## 10.1 Demand identity

The equilibrium allocation equation can be rearranged as

$$T(b(p, N), p) = \frac{pZ}{Nb(p, N)} - \frac{I}{N}. \quad (18)$$

Differentiating with respect to  $p$  gives

$$\frac{d}{dp} T(b(p, N), p) = \frac{Z}{Nb(p, N)} \left( 1 - p \frac{d}{dp} \ln b(p, N) \right). \quad (19)$$



## 10.2 Theorem 2

For large markets and any price below  $\bar{v}$ ,

$$\frac{d}{dp}T(b(p, N), p) < 0. \quad (20)$$

Thus, as the offer price falls, uninformed demand rises.

**Economic meaning.** There are two opposing effects of a lower price. A lower price raises expected returns, which raises uninformed demand. But a lower price also lets informed investors buy a larger fraction of good issues, worsening rationing. Theorem 2 says that in large markets the price effect dominates: lowering the price increases uninformed participation.

## 10.3 Full-subscription price

The full-subscription price  $p_f$  is the price at which uninformed demand alone is sufficient to purchase the entire issue in the bad state:

$$NT(b(p_f, N), p_f) = p_f Z. \quad (21)$$

Substituting into the good-state allocation equation gives

$$b(p_f, N) = \frac{p_f Z}{p_f Z + I}. \quad (22)$$

For large markets this is approximated by

$$b_0(p_f) = \frac{p_f Z}{p_f Z + I}. \quad (23)$$

If  $I = \bar{v}Z$ , then

$$b_0(p_f) = \frac{p_f}{p_f + \bar{v}}. \quad (24)$$

**Economic meaning.** The full-subscription price is more conservative than the price merely needed to generate some uninformed participation. It is the price at which the issuer can count on selling the issue even when informed investors do not buy.

# 11 Example: uniform values and quadratic utility

## 11.1 Assumptions

Rock illustrates the model with

$$\tilde{v} \sim U(0, 2\bar{v}), \quad (25)$$

and identical quadratic utility for uninformed investors,

$$U(w) = w - \frac{g}{2}w^2. \quad (26)$$

## 11.2 Uninformed demand

The first-order condition has a closed-form solution:

$$T(b/b', p) = 3 \frac{1-g}{g} \left\{ \frac{(b/b')(2\bar{v}/p - 1)^2 - 1}{(b/b')(2\bar{v}/p - 1)^3 + 1} \right\}. \quad (27)$$

This expression shows directly that uninformed demand rises when the issue is cheaper and when the allocation bias  $b/b'$  becomes less severe.

## 11.3 Zero-demand probability under the uniform distribution

Using equation (13):

$$\mathbb{P}(\tilde{v} < p) = \frac{p}{2\bar{v}}, \quad (28)$$

$$\mathbb{P}(\tilde{v} > p) = 1 - \frac{p}{2\bar{v}}, \quad (29)$$

$$\mathbb{E}[p - \tilde{v} \mid \tilde{v} < p] = \frac{p}{2}, \quad (30)$$

$$\mathbb{E}[\tilde{v} - p \mid \tilde{v} > p] = \frac{2\bar{v} - p}{2}. \quad (31)$$

Therefore,

$$b_0(p) = \frac{p^2}{(2\bar{v} - p)^2} = \frac{1}{(2\bar{v}/p - 1)^2}. \quad (32)$$

## 11.4 Full-subscription price in the large-market example

If  $I = \bar{v}Z$ , equation (23) becomes

$$\frac{1}{(2\bar{v}/p - 1)^2} = \frac{p}{p + \bar{v}}. \quad (33)$$

Let  $x = p/\bar{v}$ . Then

$$\frac{x^2}{(2 - x)^2} = \frac{x}{1 + x}. \quad (34)$$

For  $x > 0$ ,

$$x(1 + x) = (2 - x)^2 = 4 - 4x + x^2, \quad (35)$$

so

$$5x = 4, \quad x = 0.8. \quad (36)$$

The asymptotic full-subscription price is therefore

$$p_f = 0.8\bar{v}. \quad (37)$$

This corresponds to a 20 percent discount from mean value.

## 11.5 Finite-market comparison

In the numerical example, the large but finite market requires a lower price for full subscription. Table 2 reports that full subscription occurs at about 74 percent of mean value rather than exactly 80 percent. This gap reflects finite-market effects: individual orders are not infinitesimal, so the large-market zero-demand approximation is not exact.

## 12 Multiple allocation equilibria and hot/cold markets

### 12.1 Relative allocation probability

Once bad issues are also oversubscribed,  $b' < 1$ , and the relevant object is the ratio

$$\frac{b}{b'}. \quad (38)$$

At prices below the full-subscription point, both good and bad issues are rationed. The bias against uninformed investors can shrink because the bad state is now rationed too.

### 12.2 Demand explosion below full subscription

When price falls below  $p_f$ , uninformed demand can jump upward. The logic is self-reinforcing:

1. lower price increases uninformed expected returns;
2. bad-state rationing begins;
3. the relative bias  $b/b'$  improves;
4. uninformed demand increases further;
5. higher uninformed demand makes rationing in both states more similar;
6. the bias falls again and demand increases more.

This feedback creates a discontinuity in the example, visible as the jump from point B to point C in Figure 2.

### 12.3 Hot and cold issue interpretation

Rock speculates that the lower branch of the relative-allocation curve corresponds to cold markets: few offerings, large discounts, and some undersubscription. The upper branch corresponds to hot markets: heavy demand, full subscription, and smaller discounts. This interpretation is tentative, but it shows how a static IPO allocation model can generate multiple market regimes.

## 13 Issuer's optimal offer price

### 13.1 Issuer objective

The issuer or original owners are assumed to be risk averse. For illustration, Rock uses

$$U_o(W) = -W^{-1}. \quad (39)$$

The issuer values the offering because it provides risk sharing and liquidity: owners can sell shares rather than remain exposed to the entire value of the company.

### 13.2 Issuer terminal wealth

- If  $\tilde{v} > p$ , the issue is fully subscribed and the issuer receives  $pZ$ .

- If  $\tilde{v} < p$ , informed investors do not participate; the issuer sells only to uninformed investors and retains the unsold shares. Terminal wealth equals

$$NT(b, p) + \tilde{v} \left( Z - \frac{NT(b, p)}{p} \right), \quad (40)$$

where  $NT(b, p)/p$  is the number of shares sold to the uninformed.

### 13.3 Why the issuer will not choose zero uninformed demand

If uninformed demand is zero, then in bad states no shares are sold and the issuer retains the issue. Under the illustrative utility function, the possibility of very low terminal wealth is extremely costly. Hence the issuer willingly underprices to attract uninformed participation and reduce downside risk.

### 13.4 Optimal price in the example

For the uniform example, Rock shows that the constrained maximization solution occurs at the boundary:

$$p = 0.8\bar{v}. \quad (41)$$

Thus the issuer chooses the smallest discount that guarantees full subscription in every state in the large-market approximation.

**Economic intuition.** Once the price is low enough to ensure full subscription, further discounts provide little additional insurance but reduce proceeds. The optimal offer price is therefore the least discount that gives the issuer the desired risk-sharing benefit.

## 14 Empirical implications and tests

- **Allocation-adjusted returns.** If the model is correct, IPO returns weighted by allocation probabilities should approximately equal the risk-free rate for uninformed investors.
- **Rationing and returns.** The most underpriced IPOs should be the most rationed. This explains why reported first-day IPO returns are difficult for uninformed investors to capture.
- **Overallotment options.** If overallotment options are used frequently, this is indirect evidence that oversubscription is common.
- **Bias test.** IPOs for which overallotment is exercised should appreciate more than IPOs where the option expires unused, because exercise indicates stronger demand and likely good-state rationing.
- **Tender-offer analogy.** The model predicts the reverse logic for tender offers: uninformed shareholders tender good shares when the price is too high and retain bad shares when it is too low. Hence acquirers must offer a premium to overcome an analogous winner's curse.
- **Underwriting discretion.** If underwriters can identify uninformed investors and allocate more to them, required underpricing should fall. If underwriters favor informed investors, underpricing should rise.
- **Uncertainty implication.** The greater the uncertainty about true issue value, the larger the informed advantage and the larger the required discount.

## 15 Tables and figures

### 15.1 Table 1: Terminal wealth of the uninformed investor

Read:

- Table 1 shows terminal wealth for an uninformed investor as a function of two events: whether the issue is underpriced or overpriced, and whether the investor receives an allocation.
- If the issue is underpriced and the investor gets shares, wealth is  $p^{-1}\tilde{v}T + (1 - T)$ ; if not allocated, wealth is 1.
- If the issue is overpriced and allocated, the same payoff formula applies but the return is negative on average.
- The crucial asymmetry is in the probabilities:  $b_e$  applies in the underpriced state, while  $b'_e$  applies in the overpriced state.

**Economic message:** The table is the direct source of the uninformed investor's expected utility. It makes the winner's curse transparent: allocations are less likely when shares are valuable and more likely when shares are unattractive.

### 15.2 Table 2: Uninformed demand and allocation probabilities

Read:

- Table 2 reports the numerical example under large and small market assumptions.
- As the offer price falls from 95 percent to 50 percent of mean value, uninformed demand rises sharply.
- At the same time, the probability  $b$  of receiving an allocation of underpriced shares falls.
- The zero-demand probability  $b_0(p)$  also falls as the discount deepens.
- In the large-market example, the full-subscription point occurs around 74 percent of mean value, compared with the asymptotic 80 percent calculation.

**Economic message:** Lower prices attract uninformed demand despite worse good-state rationing. Underpricing therefore expands the market's ability to absorb the issue, but it also increases the informed investors' crowding-in advantage.

Table 1  
Terminal wealth of investor as a function of the aftermarket value of the new issue and the probability of obtaining an allocation.

|             | Aftermarket value <sup>a</sup>   |                             |                                 |                              |
|-------------|----------------------------------|-----------------------------|---------------------------------|------------------------------|
|             | $\tilde{v} > p$<br>(underpriced) |                             | $\tilde{v} < p$<br>(overpriced) |                              |
|             | yes                              | no                          | yes                             | no                           |
| Wealth      | $p^{-1}\tilde{v}T + (1 - T)$     | 1                           | $p^{-1}\tilde{v}T + (1 - T)$    | 1                            |
| Probability | $b_e p(\tilde{v} > p)$           | $(1 - b_e)p(\tilde{v} > p)$ | $b'_e p(\tilde{v} < p)$         | $(1 - b'_e)p(\tilde{v} < p)$ |

<sup>a</sup>Aftermarket value is the price,  $v$ , realized on the first trade; the aftermarket price differs from the offering price,  $p$ , according to whether the issue is underpriced ( $v > p$ ) or overpriced ( $v < p$ ). The probabilities of these two events from the viewpoint of the uninformed investor are denoted  $p(v > p)$  and  $p(v < p)$ , respectively. Given the issue is underpriced, the probability of an allocation is  $b_e$ ; given the issue is overpriced, the probability of an allocation is  $b'_e$ . The uninformed investor has unit wealth initially, and chooses a fraction,  $T$ , to invest in the new issue.

(a) Table 1: Terminal wealth of uninformed investor.

Table 2  
Uninformed demand and probability of an allocation of underpriced shares to an uninformed investor, as a function of offer price and number of investors; large (A) and small (B) markets.

| Offer price<br>as percentage of<br>mean price/share<br>$100 \times p/\bar{v}$ | Uninformed demand<br>as percentage of<br>mean value of issue<br>$100 \times NT^*/\bar{v}Z$ |     | Uninformed demand<br>as percentage of<br>market value of issue<br>$100 \times NT^*/\bar{p}Z$ |     | Probability of<br>an allocation of<br>underpriced shares<br>$b$ |     | Zero<br>demand<br>probability<br>$b_0(p)$ |
|-------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|-----|----------------------------------------------------------------------------------------------|-----|-----------------------------------------------------------------|-----|-------------------------------------------|
|                                                                               |                                                                                            |     |                                                                                              |     |                                                                 |     |                                           |
|                                                                               | (A)                                                                                        | (B) | (A)                                                                                          | (B) | (A)                                                             | (B) |                                           |
| 95%                                                                           | 14%                                                                                        | 17% | 15%                                                                                          | 18% | 84%                                                             | 86% | 82%                                       |
| 90                                                                            | 27                                                                                         | 25  | 30                                                                                           | 28  | 71                                                              | 72  | 67                                        |
| 85                                                                            | 41                                                                                         | 30  | 50                                                                                           | 35  | 59                                                              | 66  | 55                                        |
| 80                                                                            | 56                                                                                         | 34  | 70                                                                                           | 43  | 51                                                              | 60  | 44                                        |
| 75                                                                            | 71                                                                                         | 40  | 96                                                                                           | 53  | 43                                                              | 53  | 36                                        |
| 70                                                                            | 86                                                                                         | 47  | 123                                                                                          | 67  | 38                                                              | 47  | 29                                        |
| 65                                                                            | 100                                                                                        | 55  | 154                                                                                          | 85  | 33                                                              | 42  | 23                                        |
| 60                                                                            | 114                                                                                        | 62  | 190                                                                                          | 103 | 29                                                              | 37  | 18                                        |
| 55                                                                            | 129                                                                                        | 71  | 235                                                                                          | 129 | 25                                                              | 32  | 14                                        |
| 50                                                                            | 143                                                                                        | 80  | 286                                                                                          | 160 | 21                                                              | 28  | 11                                        |

(b) Table 2: Demand and allocation probabilities.

### 15.3 Figure 1: Probability of receiving underpriced shares

Read:

- Figure 1 plots the probability  $b$  of receiving an allocation of underpriced shares as a function of the offer price relative to mean value.
- On the solid segment, bad shares are not rationed; only good shares are rationed.
- Point B is the full-subscription boundary in the finite example, around  $p/\bar{v} = 0.74$ .
- The dotted part below B is not consistent with the assumption that bad shares are never rationed.

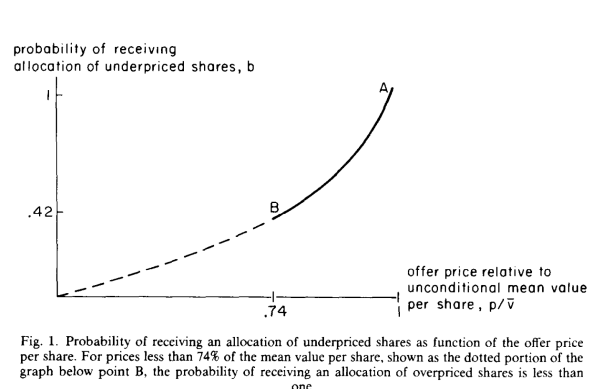
**Economic message:** As the discount deepens, informed investors can buy a larger fraction of good issues, so the probability of an uninformed allocation in good states falls. The curve visualizes the allocation side of the winner's curse.

### 15.4 Figure 2: Relative allocation probability and multiple regimes

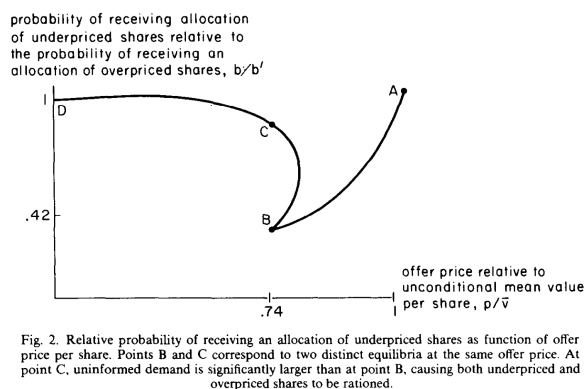
Read:

- Figure 2 plots the relative allocation probability  $b/b'$ .
- The segment A-B is the same as in Figure 1 because  $b' = 1$  there.
- At prices below the full-subscription point, both good and bad states are rationed, so  $b'$  falls below one.
- Points B and C represent two distinct equilibria at roughly the same offer price: a low-demand branch and a high-demand branch.

**Economic message:** Once both states are rationed, the allocation bias can shrink and uninformed demand can jump. This helps explain why IPO markets may exhibit discontinuous hot/cold behavior.



(a) Figure 1: Good-state allocation probability.



(b) Figure 2: Relative allocation probability.

### 15.5 Table 3: Issuer terminal wealth

Read:

- Table 3 gives the issuer's terminal wealth when the aftermarket value is above or below the offer price.
- If the issue is underpriced, it is fully subscribed and the issuer receives  $pZ$ .
- If the issue is overpriced, the issuer sells only the shares demanded by uninformed investors and retains the rest.
- The issuer's retained shares are worth  $\tilde{v}(Z - NT(b, p)/p)$  in the bad state.

**Economic message:** The table shows why risk-averse owners value uninformed participation. A lower offer price can reduce the downside risk of being stuck with unsold shares in bad states.

Table 3  
Issuer's terminal wealth, as a function of the realized value of the shares in the aftermarket.

|                 | Aftermarket value <sup>a</sup>   |                                        |
|-----------------|----------------------------------|----------------------------------------|
|                 | $\tilde{v} > p$<br>(underpriced) | $\tilde{v} < p$<br>(overpriced)        |
| Issuer's wealth | $pZ$                             | $NT(b, p) + \tilde{v}(Z - NT(b, p)/p)$ |

<sup>a</sup>Aftermarket value is the price,  $v$ , realized on the first trade; this price differs from the offering price,  $p$ , according to whether the issue is underpriced ( $v > p$ ) or overpriced ( $v < p$ ). The number of shares offered is  $Z$ , the number of uninformed investors is  $N$ , and  $T$  is the fraction of wealth the uninformed investor allocates to the new issue.

Table 3: Issuer terminal wealth.

## 16 Short summary

- Rock explains IPO underpricing as compensation for a winner's curse in allocations.
- Informed investors know whether the issue is good and participate only when  $\tilde{v} > p$ .
- Good issues attract informed demand and become more rationed; bad issues do not attract informed demand and are easier for the uninformed to obtain.
- Uninformed investors therefore demand a discount to participate.
- The key state variable for uninformed demand is the relative rationing probability  $b/b'$ .
- In large markets, the equilibrium allocation probability converges to a zero-demand probability  $b_0(p)$  that makes uninformed investors just willing to participate.
- In the uniform example,  $b_0(p) = 1/(2\bar{v}/p - 1)^2$ .
- The full-subscription price solves  $b_0(p) = pZ/(pZ + I)$ ; when  $I = \bar{v}Z$ , the asymptotic solution is  $p = 0.8\bar{v}$ .
- With risk-averse owners, the issuer chooses a discount to obtain insurance against unsold shares in bad states.
- The model predicts that raw IPO returns overstate the gains available to ordinary uninformed investors because underpriced issues are disproportionately rationed.

## **17 Conclusion**

### **17.1 Core conclusion**

Rock provides a canonical winner's-curse theory of IPO underpricing. Because informed investors selectively demand good issues, uninformed investors receive biased allocations. IPOs must be priced below expected aftermarket value to compensate uninformed investors for this adverse allocation.

### **17.2 Economic intuition**

The key event is not the offer price alone but the conditional meaning of receiving shares. When an uninformed investor receives a large allocation, this may indicate that informed investors were not interested. Thus, the allocation itself contains bad news. Underpricing restores participation by making even this biased allocation acceptable.

### **17.3 Incremental contribution revisited**

The paper turns IPO underpricing from an unexplained empirical regularity into an equilibrium outcome of asymmetric information and rationing. Its central contribution is to connect the institutional allocation mechanism of IPOs to the information structure of investor demand.

### **17.4 Implications for issuers and underwriters**

Issuers benefit from uninformed participation because it makes proceeds more reliable. Underwriters matter because their allocation rules determine the severity of the winner's curse. Allocation practices that favor informed investors require deeper discounts; allocation practices that protect uninformed investors can reduce underpricing.

### **17.5 Empirical implications**

The cleanest test is not whether IPOs earn positive first-day returns. The model predicts that they should. The sharper test is whether allocation-adjusted returns for uninformed investors are approximately normal. Underpriced issues should be rationed more heavily than overpriced issues.

### **17.6 Limitations**

The model abstracts from underwriter certification, bookbuilding, price revision, strategic issuers, repeated investor-underwriter relations, and aftermarket price support. It also assumes extreme informed investors with perfect information. These simplifications make the mechanism stark but not exhaustive.

### **17.7 Takeaway**

IPO underpricing is the equilibrium cost of selling new shares to a mixed investor pool in which some investors know when to show up and uninformed investors know they will be crowded out precisely when showing up is most valuable.